Stylized Facts and Factor Exposures of JPMorgan Chase Stock Returns, 1980–2025

Empirical study

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Abstract

This study examines 45 years of daily, weekly, monthly and annual returns of JPMorgan Chase & Co. (NYSE: JPM). The first part tests seven stylized facts of financial returns: stationarity, negative skewness, heavy tails, aggregational Gaussianity, absence of linear autocorrelation, volatility clustering and the leverage effect. Standard tests assume i.i.d. normal returns, which the data reject, so each one is paired with a procedure that remains valid under heavy tails and conditional heteroskedasticity: block homogeneity tests, the Hill estimator, Student-t maximum likelihood, White heteroskedasticity-consistent standard errors, Engle's ARCH test and a GARCH(1,1) model estimated by maximum likelihood. Heavy tails (tail index close to 3) and volatility clustering (GARCH persistence 0.993) hold strongly. Negative skewness holds at weekly to monthly horizons, but not at the daily frequency, where it is produced by the October 1987 crash. The apparent autocorrelation of daily and weekly returns, and much of the statistical evidence for a leverage effect in JPM's own returns, disappear once heteroskedasticity is accounted for. The second part estimates the CAPM and the Fama–French three-factor model on monthly excess returns. JPM has a market beta significantly above one (1.24 in the CAPM, 1.35 with three factors), a strong value exposure (HML loading 0.87), no size exposure, and no significant alpha in either model.

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1 Introduction

Empirical finance has documented a set of statistical properties shared by the returns of most liquid assets, often called stylized facts: returns are close to unpredictable, but their distribution has heavy tails, their volatility is clustered and persistent, and large falls tend to be followed by higher volatility. These properties matter in practice: they determine how risk should be measured, which models are appropriate for volatility and option pricing, and how much confidence can be placed in standard statistical tests.

This study tests these facts on a single, large and highly cyclical stock, JPMorgan Chase, over 1980–2025, and then measures its systematic risk exposures with the two standard linear factor models. Two methodological choices run through the analysis.

- **Robustness of inference.** Most textbook tests of stylized facts are calibrated under i.i.d. normal returns, an assumption the data reject. Each standard test is therefore complemented by a test that remains valid under heavy tails and conditional heteroskedasticity, and the conclusions are drawn from both.
- **Robustness to extreme observations.** The October 1987 crash produces the largest daily loss of the sample. Every result that may depend on a handful of observations is re-estimated without the crash week and, where relevant, by sub-period.

Section 2 describes the data and Section 3 the testing approach. Section 4 examines the seven stylized facts, Section 5 the factor models. Section 6 concludes and Section 7 discusses limitations. The appendix reports the complete autocorrelation tables.

2 Data

Table 1: Data sources.

Series	Source	Frequency	Sample
JPM adjusted closing price	Yahoo Finance	Daily	17 Mar 1980 – 31 Dec 2025
CBOE Volatility Index (VIX)	Yahoo Finance	Daily	Jan 1990 – Dec 2025
Mkt−RF, SMB, HML, RF	K. R. French Data Library	Monthly	Apr 1980 – Dec 2025

Asset and prices. JPMorgan Chase & Co. is traded on the New York Stock Exchange under the ticker JPM. The sample starts on the first date available, 17 March 1980, and ends on 31 December 2025: 45.8 years, 11,543 trading days, 550 months and 46 calendar years. Prices are daily closing prices adjusted for splits and dividends.

Returns. The first part uses log-returns $r_t = \ln P_t - \ln P_{t-1}$. Weekly, monthly and annual prices are the last observation of each calendar week, month and year, which gives 11,542 daily, 2,389 weekly, 549 monthly and 45 annual returns. Log-returns are additive over time, so a return over a longer horizon is the sum of the shorter returns that compose it. The second part uses simple returns $R_t = P_t/P_{t-1} - 1$, for the reasons given in Section 5.7.

Factors. The Fama–French file provides the excess return of the CRSP value-weighted index of all US stocks (Mkt−RF), the size factor SMB (small minus big), the value factor HML (high minus low book-to-market) and the one-month Treasury bill rate RF, all in percent per month. Only the monthly block of the file is used.

The October 1987 crash. The largest daily loss of the sample, a log-return of -32.5%, occurred in the week of 19 October 1987. Results that may depend on it are re-estimated after removing the five trading days of 19–23 October 1987 (the "ex-crash" sample, 11,537 daily returns).

3 Methodology

Standard tests and their assumption. Several standard tests of stylized facts use standard errors derived under i.i.d. normal returns:

$$z_S = \frac{\hat{S}}{\sqrt{6/N}}, \qquad z_K = \frac{\hat{K}-3}{\sqrt{24/N}}, \qquad \hat{\rho}_j \pm \frac{1.96}{\sqrt{N}}, \qquad Q_{LB}(m) = N(N+2)\sum_{j=1}^{m}\frac{\hat{\rho}_j^2}{N-j} \sim \chi^2(m).$$

When the null hypothesis is normality itself, as in the kurtosis test, this is appropriate. When it is not, the assumption matters. Under conditional heteroskedasticity, large returns occur together, and the dispersion of a sample autocorrelation exceeds $1/N$: Bartlett bands and Ljung–Box tests then reject too often. Under heavy tails, the dispersion of the sample skewness exceeds $\sqrt{6/N}$.

Robust complements. Each standard procedure is therefore paired with one that does not rely on that assumption (Table 2).

Table 2: Standard tests and robust complements used for each stylized fact.

Question	Standard procedure	Complement
Are the moments constant?	Rolling moments; linear trend on block moments	Homogeneity test across non-overlapping blocks
Is the distribution skewed?	z-test on $\hat{S}$ at four frequencies	Skewness at every horizon from 1 to 252 days, with and without the 1987 crash
How heavy are the tails?	z-test on $\hat{K}$; kurtosis-implied Student-t	Hill estimator on each tail; Student-t maximum likelihood
Do returns become normal when aggregated?	Jarque–Bera, Lilliefors, Anderson–Darling	Jarque–Bera/N, a shape measure comparable across sample sizes
Are returns autocorrelated?	Bartlett bands; Box–Pierce and Ljung–Box	Autoregression with White standard errors and robust F-test
Is volatility clustered?	ACF of r_t^2 and $\lvert r_t\rvert$; Ljung–Box on r_t^2	Engle's ARCH-LM test; GARCH(1,1) by maximum likelihood
Is there a leverage effect?	$\mathrm{corr}(r_{t-j}, r_t^2)$ with $z = \hat{\rho}_j\sqrt{N}$	Regression of r_t^2 on lagged returns with White standard errors; regressions on VIX changes

White standard errors. In a regression $y_t = x_t'\beta + u_t$, the OLS covariance matrix $\hat{\sigma}^2(X'X)^{-1}$ assumes a constant error variance. White's heteroskedasticity-consistent estimator replaces it by

$$\widehat{\mathrm{Var}}_{\mathrm{White}}(\hat{\beta}) = (X'X)^{-1}\Big(\sum_t \hat{u}_t^2\, x_t x_t'\Big)(X'X)^{-1},$$

which weights each observation by its own squared residual. The HC0 version above is used throughout; the HC3 small-sample correction is reported for the factor models. Joint hypotheses $R\beta = r$ are tested with the Wald F-statistic computed with either covariance matrix. If returns were homoskedastic, the OLS and White columns of the tables would coincide; the gap between them measures the effect of heteroskedasticity on standard inference.

All tests are run at the 5% level. Means, standard deviations and quantiles are in percent unless stated otherwise.

4 Stylized facts of JPM returns

4.1 Summary statistics

Table 3 gives the main statistics. The average daily return is 0.050%, and its 95% confidence interval, $[0.010\%, 0.090\%]$, excludes zero. The standard deviation grows roughly with the square root of the horizon, from 2.2% per day to 9.0% per month and 30.4% per year. All frequencies show negative skewness and a kurtosis above 3; the kurtosis declines with the horizon. Normality is rejected everywhere by Jarque–Bera and at every frequency but the annual one by Lilliefors.

The four sub-periods of Table 4 are of equal length (2,885 or 2,886 observations). They differ strongly in volatility (from 1.69% to 2.60% per day), in skewness (-1.31 in 1980–1991, positive in the two middle periods) and in kurtosis (from 7.4 to 28.2), but much less in the mean. These differences are examined below.

Table 3: Summary statistics of JPM log-returns by frequency. "Daily, ex-crash" removes 19–23 October 1987. The C.I. half-width is $1.96\,\hat{\sigma}/\sqrt{N}$. Lilliefors p-values below 0.001 are at the lower bound of the tabulated distribution.

	Daily	Daily, ex-crash	Weekly	Monthly	Annual
Mean (%)	0.050	0.052	0.238	1.040	12.238
Standard deviation (%)	2.205	2.182	4.746	9.036	30.422
95% C.I. half-width of the mean (%)	0.040	0.040	0.190	0.756	8.889
Skewness	−0.106	0.166	−0.210	−0.743	−0.964
Kurtosis	18.29	14.76	11.81	5.55	4.90
Excess kurtosis	15.29	11.76	8.81	2.55	1.90
Minimum (%)	−32.46	−23.23	−41.68	−42.58	−92.36
5% quantile (%)	−3.15	−3.15	−7.28	−13.94	−35.21
25% quantile (%)	−0.94	−0.94	−2.06	−3.62	−5.69
Median (%)	0.00	0.00	0.30	1.41	11.05
75% quantile (%)	1.03	1.03	2.70	6.62	30.88
95% quantile (%)	3.24	3.24	7.26	14.78	51.36
Maximum (%)	22.39	22.39	39.94	28.45	73.04
Jarque–Bera statistic	112,450.5	66,481.3	7,739.0	199.3	13.7
Jarque–Bera p-value	<0.001	<0.001	<0.001	<0.001	0.001
Lilliefors statistic	0.084	0.083	0.066	0.068	0.123
Lilliefors p-value	<0.001	<0.001	<0.001	<0.001	0.090
Observations	11,542	11,537	2,389	549	45

4.2 Stylized fact 1: returns are stationary

Prices versus returns. The price and the log-price trend upwards (Figure 1). The autocorrelation function of the log-price decays extremely slowly: it is still 0.90 after 250 trading days at the daily frequency, fifty times the Bartlett half-width, and above 0.70 after 40 months at the monthly frequency (Figure 2, Table 5). This is the signature of a non-stationary, integrated series. Returns, by contrast, fluctuate around a constant level, and their autocorrelations are inside or at the edge of the band at every lag.

Rolling moments. The mean and standard deviation are computed over rolling windows of 252 days (Figure 3). The band around the rolling mean is $\bar{r}_t \pm 1.96\,\hat{\sigma}_t/\sqrt{252}$; the band around the rolling standard deviation uses the asymptotic standard error $\sqrt{\mu_4 - \sigma^4}/(2\sigma\sqrt{252})$, which accounts for the fourth moment. The rolling mean oscillates around its full-sample value and its

Table 4: Summary statistics of JPM daily log-returns over four sub-periods of equal length.

	1980–1991	1991–2003	2003–2014	2014–2025
Mean (%)	0.042	0.044	0.041	0.071
Standard deviation (%)	2.055	2.369	2.599	1.688
95% C.I. half-width of the mean (%)	0.075	0.086	0.095	0.062
Skewness	−1.311	0.126	0.328	−0.083
Kurtosis	28.16	7.41	18.88	16.14
Excess kurtosis	25.16	4.41	15.88	13.14
Minimum (%)	−32.46	−19.98	−23.23	−16.21
5% quantile (%)	−2.94	−3.62	−3.56	−2.52
25% quantile (%)	−0.99	−1.28	−0.90	−0.70
Median (%)	0.00	0.00	0.03	0.07
75% quantile (%)	1.06	1.25	0.97	0.87
95% quantile (%)	3.24	3.77	3.32	2.39
Maximum (%)	12.89	14.87	22.39	16.56
Jarque–Bera statistic	76,898.5	2,350.1	30,354.7	20,751.1
Jarque–Bera p-value	<0.001	<0.001	<0.001	<0.001
Lilliefors statistic	0.068	0.057	0.127	0.085
Lilliefors p-value	<0.001	<0.001	<0.001	<0.001
Observations	2,885	2,886	2,885	2,886

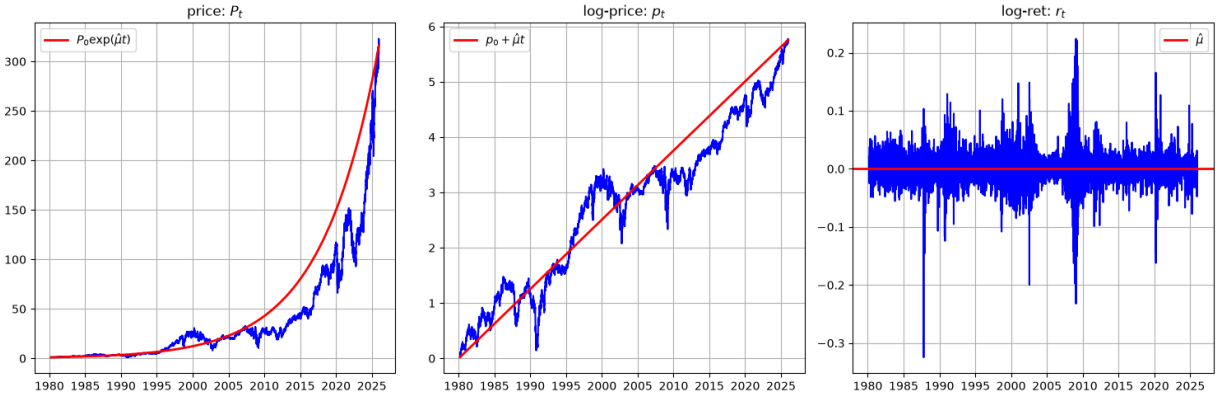


Figure 1: JPM daily price, log-price and log-return, with the fitted trends $P_0 e^{\hat{\mu}t}$, $p_0 + \hat{\mu}t$ and $\hat{\mu}$ (red).

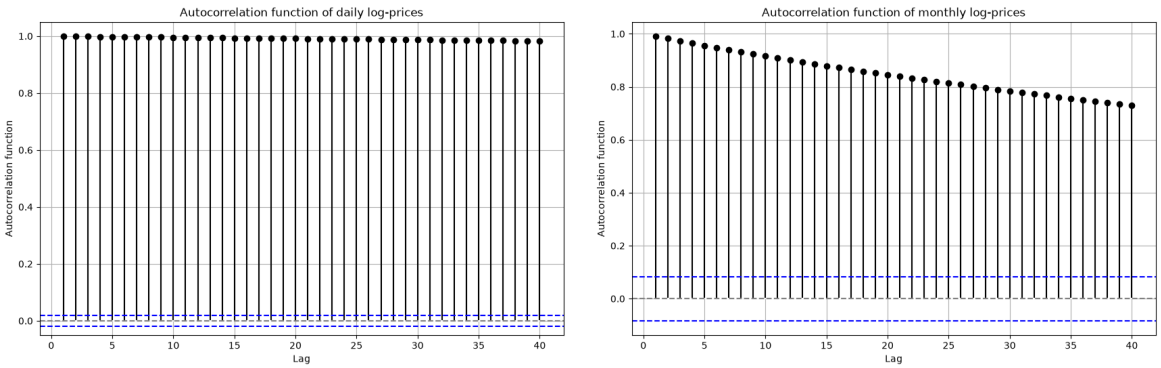


Figure 2: Autocorrelation function of daily (left) and monthly (right) log-prices, 40 lags, with Bartlett bands $\pm 1.96/\sqrt{N}$.

Table 5: Autocorrelation function of daily log-prices and daily log-returns at selected lags.

Lag	Daily log-price	Daily log-return
1	0.9995	−0.0323
5	0.9978	0.0060
20	0.9913	0.0239
60	0.9745	0.0068
125	0.9485	0.0009
250	0.9025	0.0039
Bartlett half-width $1.96/\sqrt{N}$	0.0182	0.0182

band almost always contains it. The rolling standard deviation moves from about 1% to almost 7% in 2008–2009, far outside its own band.

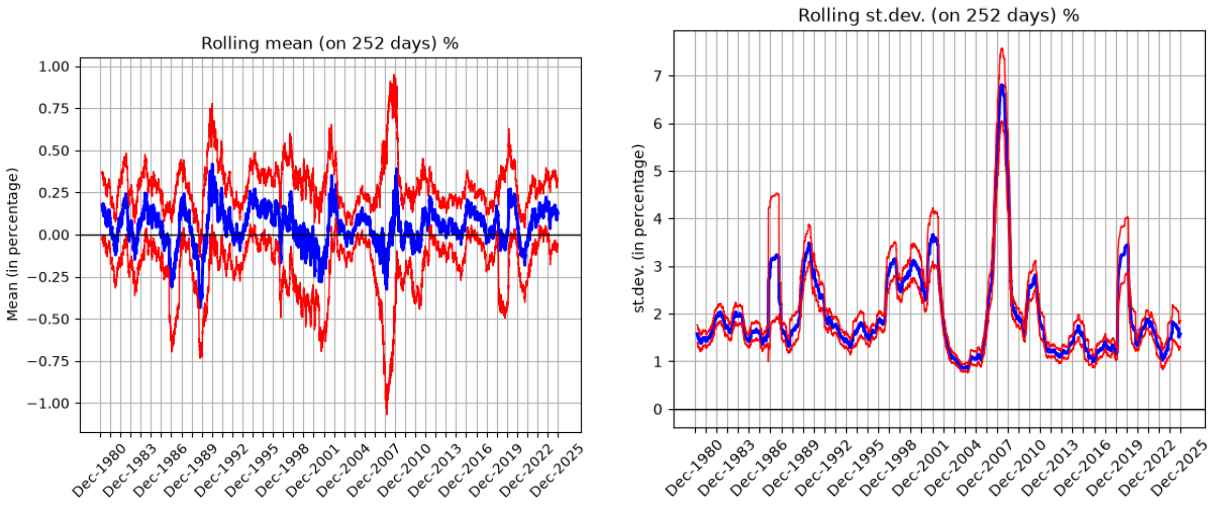


Figure 3: Rolling mean (left) and rolling standard deviation (right) of daily log-returns over 252 days, in percent, with 95% confidence bands (red).

Block moments and trends. The sample is split into non-overlapping blocks: 230 blocks of 50 days, and 45 blocks of 12 months. In each block, the mean, the variance and the lag-1 autocovariance of returns are computed, and a linear trend is fitted on each series of block estimates (Figures 4 and 5, Table 6). The slope of the block means is 0.0003% per year on daily data ($t = 0.18$) and 0.0015% on monthly data ($t = 0.05$): there is no trend in the mean. The block variances show no drift either: they rise sharply in 1987, 1990–1991, 1998–2002, 2008–2009 and 2020, and return to the same level after each episode.

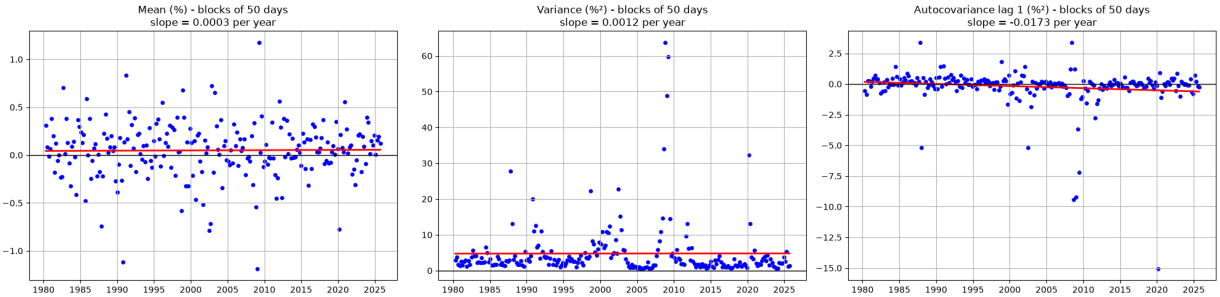


Figure 4: Mean, variance and lag-1 autocovariance of daily log-returns on 230 non-overlapping blocks of 50 days, with fitted linear trend (red).

Homogeneity test. To test formally whether a moment depends on time, the block estimates

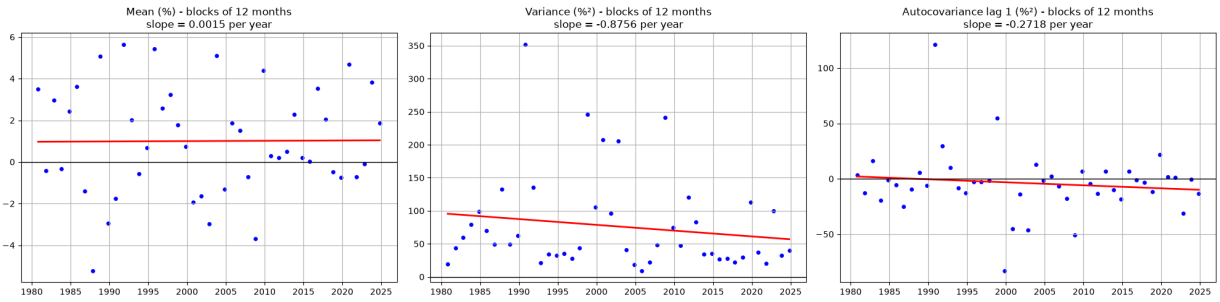


Figure 5: Mean, variance and lag-1 autocovariance of monthly log-returns on 45 non-overlapping blocks of 12 months, with fitted linear trend (red).

Table 6: Slope per year of a linear trend fitted on the block moments. The t-statistic and p-value test H_0: slope $= 0$ for the mean.

	Mean: slope	t	p-value	Variance: slope	Autocov.: slope
Daily, 50-day blocks	0.0003	0.18	0.854	0.0012	−0.0173
Monthly, 12-month blocks	0.0015	0.05	0.960	−0.8756	−0.2718

$\hat{\theta}_i$ ($i = 1, \ldots, B$, each computed on n observations v_t: r_t for the mean, $(r_t - \bar{r})^2$ for the variance, $\hat{\varepsilon}_t\hat{\varepsilon}_{t-1}$ for the autocovariance) are compared with

$$Q = \frac{n\sum_{i=1}^{B}(\hat{\theta}_i - \bar{\theta})^2}{\hat{\sigma}^2_{\text{within}}} \overset{H_0}{\approx} \chi^2(B-1), \qquad \hat{\sigma}^2_{\text{within}} = \frac{1}{nB - B}\sum_{i=1}^{B}\sum_{t\in i}(v_t - \hat{\theta}_i)^2.$$

The denominator is the pooled variance within blocks, so the test does not assume a known variance. $Q/(B-1)$ is exactly the global F-statistic of a regression of v_t on block dummies.

Table 7: Homogeneity test of the mean, variance and lag-1 autocovariance across non-overlapping blocks. H_0: the same moment in all blocks.

	Daily, 230 blocks of 50 days			Monthly, 45 blocks of 12 months		
	Mean	Variance	Autocov.	Mean	Variance	Autocov.
Q	201.7	2,017.2	243.1	42.5	108.8	40.0
Degrees of freedom	229	229	229	44	44	44
Q/d.f.	0.88	8.81	1.06	0.97	2.47	0.91
5% critical value	265.3	265.3	265.3	60.5	60.5	60.5
p-value	0.903	<0.001	0.249	0.535	<0.001	0.646

Results. Table 7 gives a clear split between the moments.

- **Mean.** Equality across blocks is not rejected, neither with 230 blocks of 50 days ($p = 0.90$) nor with 45 blocks of 12 months ($p = 0.53$). The sub-period means of Table 4 have overlapping confidence intervals, for example $[-0.05\%, 0.14\%]$ for 2003–2014 and $[0.01\%, 0.13\%]$ for 2014–2025.
- **Autocovariance.** Equality is not rejected either ($p = 0.25$ and 0.65).
- **Variance.** Equality is strongly rejected (Q/d.f. $= 8.8$ on daily blocks, 2.5 on monthly blocks).

The rejection for the variance does not contradict stationarity. The block test detects the clustering of the conditional variance (stylized fact 6): a block that falls in 2008 has a high variance because volatility is persistent, not because the unconditional variance has changed. A constant

unconditional variance is compatible with this behaviour, and the return of block variances to the same level after each episode is what a stationary but persistent conditional variance produces. A block test cannot separate the two cases; the GARCH model of Section 4.7 does, and its persistence is below one.

Conclusion. Stationarity holds for the mean and the autocovariance. For the variance the evidence is compatible with it, the time variation being that of the conditional variance.

4.3 Stylized fact 2: negative skewness

Method. Rather than four calendar frequencies, the skewness is computed at every horizon h from 1 to 252 trading days. For a given h, the sample can be cut into non-overlapping h-day returns in h different ways, depending on the starting day. The h skewness estimates are averaged, so that the result does not depend on an arbitrary starting date. Under normality $\hat{S} \sim \mathcal{N}(0, 6/N)$ approximately, with $N = \lfloor 11{,}542/h \rfloor$, and the one-sided test H_0: $S = 0$ against H_1: $S < 0$ rejects when $z = \hat{S}/\sqrt{6/N} < -1.645$. The critical value $-1.645\sqrt{6/N}$ widens as h grows because N shrinks. The whole exercise is repeated on the price series rebuilt without the crash week.



Figure 6: Skewness of h-day log-returns for $h = 1, \ldots, 252$ trading days (average over the h possible starting days), with and without the October 1987 crash, and 5% critical value of the one-sided test.

Table 8: Skewness at the daily, weekly, monthly and annual horizons (h in trading days), one-sided test H_0: $S = 0$ vs. H_1: $S < 0$. The $h = 5$ row uses 5-trading-day returns and differs slightly from the calendar-week column of Table 3.

			With the 1987 crash			Without the 1987 crash		
	h	N	$\hat{S}$	z	p-value	$\hat{S}$	z	p-value
Daily	1	11,542	−0.106	−4.63	<0.001	0.166	7.26	1.000
Weekly	5	2,308	−0.344	−6.75	<0.001	−0.218	−4.28	<0.001
Monthly	21	549	−0.534	−5.11	<0.001	−0.484	−4.63	<0.001
Annual	252	45	−0.453	−1.24	0.108	−0.395	−1.08	0.140

Results. With the full sample, the skewness is negative at all 252 horizons and significant at 133 of them, essentially every horizon up to six months (Figure 6, Tables 8 and 9). The picture is different at the two ends.

Table 9: Horizons at which the skewness is significantly negative ($z < -1.645$).

	With the 1987 crash	Without the 1987 crash
First h not significant	133	1
Last h significant	134	118
Number of significant h (out of 252)	133	114

- **Daily frequency.** The daily result is not robust: five observations out of 11,542 produce it. Without the crash week the daily skewness becomes positive (+0.166, $z = +7.26$). It is positive or close to zero in three of the four sub-periods (Table 4). The daily quantiles are symmetric (-3.15% at 5% against $+3.24\%$ at 95%, -0.94% against $+1.03\%$ for the quartiles), and the mean (0.050%) lies above the median (0.000%), the opposite of what a negatively skewed distribution produces.
- **Intermediate horizons.** The fact is robust. The skewness stays negative with and without the crash at 5 days (-0.34 / -0.22) and at 21 days (-0.53 / -0.48). Without the crash it is still significantly negative at 114 of the 252 horizons, all between 2 and 118 days.
- **Annual horizon.** The estimate is negative (-0.45; -0.96 with calendar years) but not significant with 45 observations.

The z-statistics must be read with caution: $\sqrt{6/N}$ is the standard error under normality, which is rejected (stylized fact 3). With heavy tails the true dispersion of $\hat{S}$ is larger, so significance is overstated, most of all at short horizons.

Conclusion. Negative skewness holds from the weekly to the monthly horizon. At the daily frequency the negative sign is entirely due to October 1987.

4.4 Stylized fact 3: heavy tails

Graphical evidence. The kernel density of daily returns is far more peaked than the normal density with the same mean and variance, with much thicker tails (Figure 7). In the QQ-plots against Student-t distributions (Figure 8), the points bend away from the line for $\nu = 10$, less for $\nu = 5$, and are closest to it for $\nu = 3$.

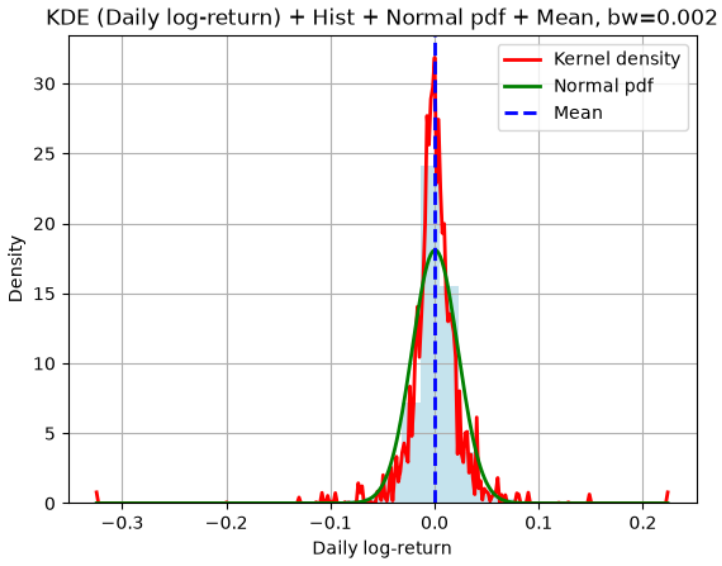


Figure 7: Histogram, kernel density estimate and normal density with the same mean and variance, daily log-returns.

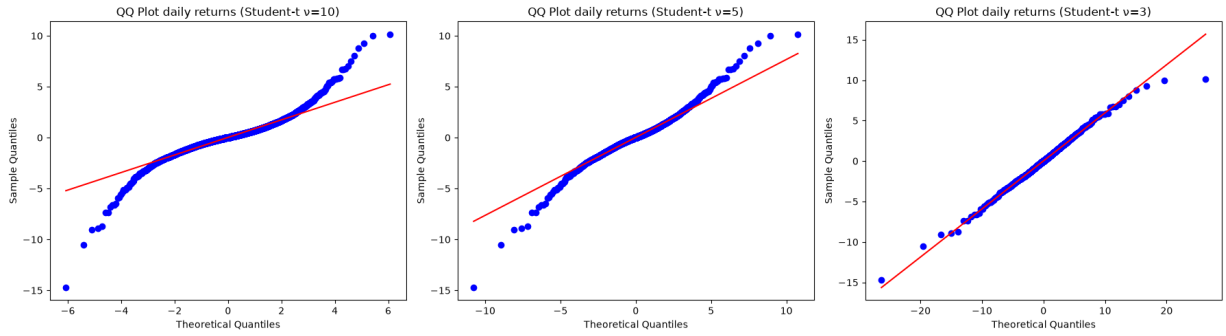


Figure 8: QQ-plots of standardized daily log-returns against Student-t distributions with $\nu = 10$, 5 and 3 degrees of freedom.

Kurtosis test. The one-sided test H_0: $K = 3$ against H_1: $K > 3$ rejects when $z = (\hat{K} - 3)/\sqrt{24/N} > 1.645$. Here the null hypothesis is normality itself, so $\sqrt{24/N}$ is the appropriate standard error. The daily kurtosis is 18.3, that is $z = 335$. Unlike the skewness, the result does not depend on the 1987 crash ($\hat{K} = 14.8$, $z = 258$ without it) and holds in every sub-period ($\hat{K} = 28.2$, 7.4, 18.9 and 16.1). Both tails are heavy: the extreme daily returns, -32.5% and $+22.4\%$, lie 14.7 and 10.1 standard deviations from the mean, events whose probability under a normal distribution is below 10^{-23}.

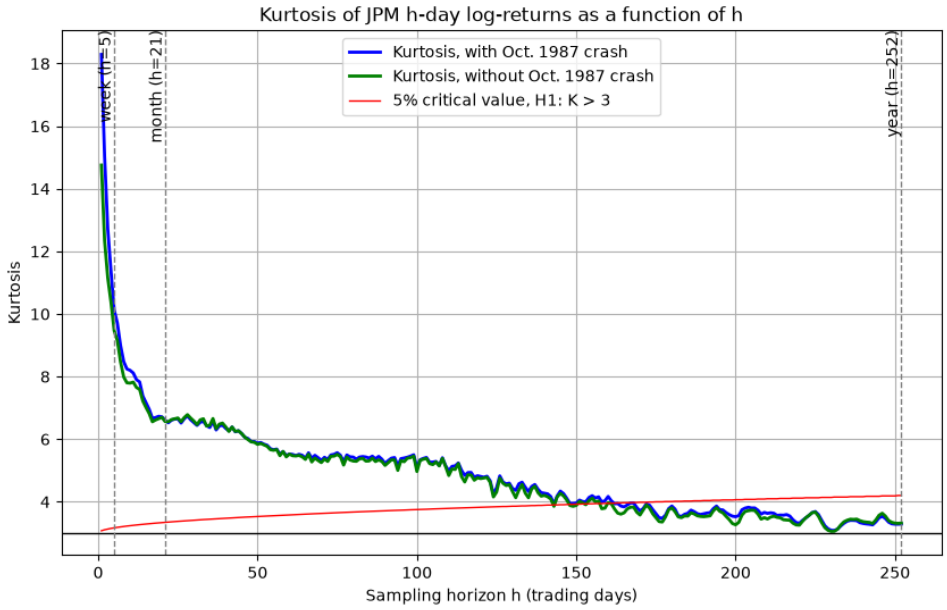


Figure 9: Kurtosis of h-day log-returns for $h = 1, \ldots, 252$ trading days, with and without the October 1987 crash, and 5% critical value of the one-sided test H_1: $K > 3$.

Table 10: Kurtosis by horizon and one-sided test H_0: $K = 3$ vs. H_1: $K > 3$. The critical kurtosis $3 + 1.645\sqrt{24/N}$ is the smallest kurtosis the test can detect at 5%. $\hat{\nu} = 4 + 6/(\hat{K} - 3)$ is the Student-t with the same kurtosis.

				With the 1987 crash				Without the 1987 crash			
	h	N	Crit. K	$\hat{K}$	z	$\hat{\nu}$	p-value	$\hat{K}$	z	$\hat{\nu}$	p-value
Daily	1	11,542	3.08	18.29	335.3	4.39	<0.001	14.76	257.7	4.51	<0.001
Weekly	5	2,308	3.17	10.14	70.0	4.84	<0.001	9.47	63.4	4.93	<0.001
Monthly	21	549	3.34	6.61	17.2	5.66	<0.001	6.56	17.0	5.68	<0.001
Annual	252	45	4.20	3.29	0.4	24.69	0.346	3.32	0.4	22.68	0.330

Table 11: Horizons at which the kurtosis is significantly above 3 ($z > 1.645$).

	With the 1987 crash	Without the 1987 crash
First h not significant	162	143
Last h significant	161	160
Number of significant h (out of 252)	161	153

How heavy are the tails? A first quantification inverts the kurtosis of a Student-t, $K = 3 + 6/(\nu - 4)$: the daily kurtosis corresponds to $\hat{\nu} = 4.4$ (Table 10). This estimator has a flaw: by construction it cannot fall below 4, the value under which the fourth moment of a Student-t is infinite, and it depends on a sample fourth moment driven by a handful of observations. Two estimators that do not rely on the fourth moment are therefore added (Table 12).

- **Hill estimator.** It uses only the k largest observations of one tail, $X_{(1)} \geq \cdots \geq X_{(k)}$, and estimates the tail index a in $P(X > x) \sim x^{-a}$ by

$$\hat{a} = \Big(\frac{1}{k}\sum_{i=1}^{k} \ln \frac{X_{(i)}}{X_{(k+1)}}\Big)^{-1}, \qquad \text{s.e.}(\hat{a}) \approx \frac{\hat{a}}{\sqrt{k}}.$$

  It is computed separately for losses and gains, with k equal to 1%, 2.5% and 5% of the sample.

- **Student-t maximum likelihood.** A Student-t distribution with location, scale and degrees of freedom is fitted to all daily returns. For a Student-t, the tail index equals ν, so the estimates are directly comparable.

Table 12: Tail index of daily log-returns: Hill estimator on the k largest observations of each tail, and Student-t degrees of freedom.

	$k = 115$ (1%)	$k = 288$ (2.5%)	$k = 577$ (5%)
Hill estimator, left tail	2.70	2.95	2.62
Hill estimator, right tail	2.97	2.73	2.47
Student-t ν, maximum likelihood	2.82		
Student-t ν, from the kurtosis	4.39		

Results. Both estimators give much heavier tails than the kurtosis suggests: the Hill estimates lie between 2.5 and 3.0 for the three values of k, and the maximum-likelihood estimate is $\hat{\nu} = 2.82$, in agreement with the QQ-plots, where $\nu = 3$ fits best. Three consequences follow.

1. With a tail index below 4, the theoretical kurtosis is infinite. This explains why the sample kurtosis is so unstable across sub-periods (from 7.4 to 28.2) and why the kurtosis-based $\hat{\nu}$ overstates the degrees of freedom.
2. With a tail index above 2, the variance is finite, as second-order stationarity requires (stylized fact 1).
3. The two tails are equally heavy: at $k = 115$, the left and right estimates (2.70 and 2.97) differ by less than their standard errors (≈ 0.26). This confirms that daily returns show no robust asymmetry (stylized fact 2).

Longer horizons. The kurtosis declines smoothly with the horizon (Figure 9). With the full sample it remains significant up to $h = 161$ days; without the crash, at 153 of the 252 horizons (Table 11). Beyond that, non-rejection reflects a loss of power rather than normality: with $N = 45$ annual returns, only a kurtosis above 4.20 can be detected.

Conclusion. Heavy tails are verified at the daily, weekly and monthly frequencies, independently of the 1987 crash, with a tail index close to 3.

4.5 Stylized fact 4: aggregational Gaussianity

Method. If returns are i.i.d. with finite variance, the central limit theorem implies that their sum over longer horizons converges to a normal distribution. The distribution of daily, weekly, monthly and annual returns is compared with the normal distribution graphically and with three tests: Jarque–Bera, which targets skewness and kurtosis; Lilliefors, a Kolmogorov–Smirnov test with estimated mean and variance; and Anderson–Darling, which gives more weight to the tails.

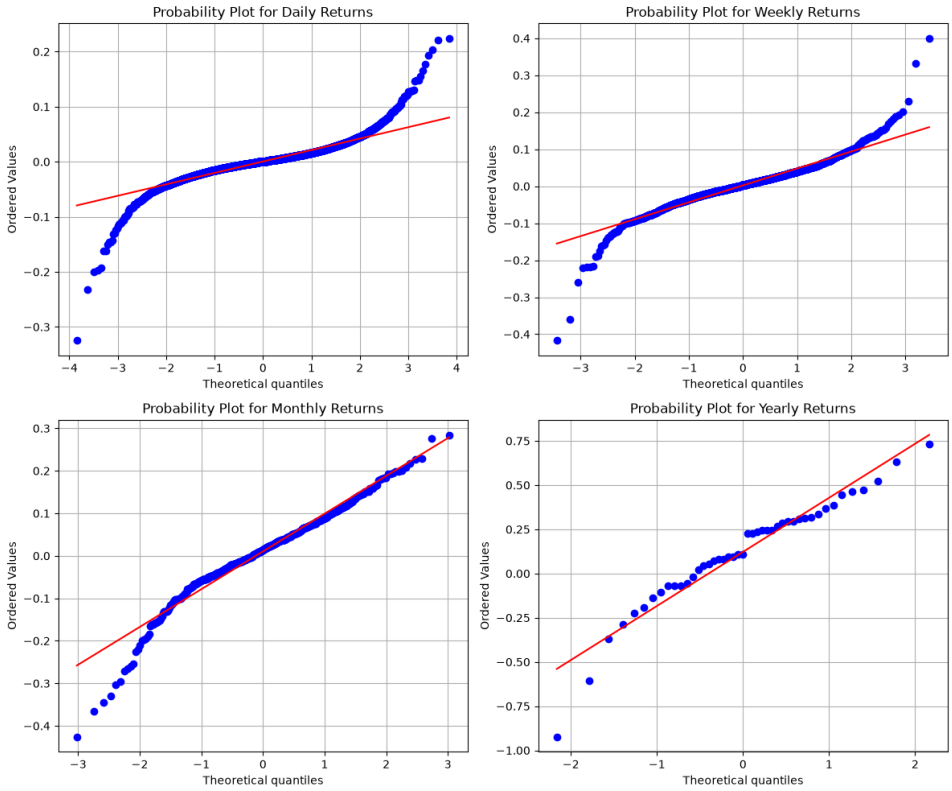


Figure 10: Normal QQ-plots of daily, weekly, monthly and annual log-returns.

Table 13: Normality tests by frequency, H_0: returns are normally distributed. $\mathrm{JB}/N = \frac{1}{6}\big(\hat{S}^2 + (\hat{K}-3)^2/4\big)$.

	Daily	Weekly	Monthly	Annual
Jarque–Bera statistic	112,450.5	7,739.0	199.3	13.7
Jarque–Bera p-value	<0.001	<0.001	<0.001	0.001
Jarque–Bera / N	9.74	3.24	0.36	0.31
Lilliefors statistic	0.084	0.066	0.068	0.123
Lilliefors p-value	<0.001	<0.001	<0.001	0.090
Anderson–Darling statistic	212.92	24.23	4.12	0.66
Anderson–Darling 5% critical value	0.752	0.752	0.751	0.739
Anderson–Darling rejects at 5%	yes	yes	yes	no
Skewness	−0.106	−0.210	−0.743	−0.964
Kurtosis	18.29	11.81	5.55	4.90

Results. The QQ-plots straighten as the horizon increases (Figure 10), the monthly histogram is much closer to the normal density than the daily one (Figure 11), and the kurtosis falls

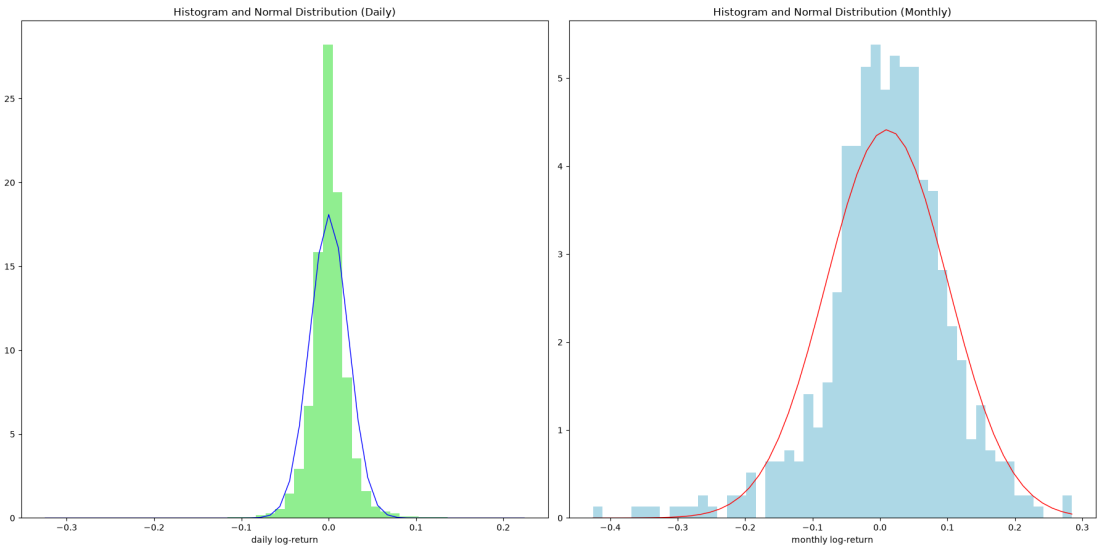


Figure 11: Histogram and normal density with the same mean and variance: daily (left) and monthly (right) log-returns.

monotonically from 18.3 (daily) to 11.8, 5.5 and 4.9 (annual). The Jarque–Bera statistic is proportional to N, which drops from 11,542 to 45, so its decline partly reflects sample size. JB/N, a pure measure of shape, is comparable across frequencies: it is divided by 27 between the daily and the monthly frequency (9.74 to 0.36), then stabilises (0.31). The implied degrees of freedom of Table 10 tell the same story ($\hat{\nu} = 4.4$, 4.8, 5.7, then 24.7 at one year).

Convergence is however incomplete. Normality is rejected by all three tests at the daily, weekly and monthly frequencies. At the annual frequency, Jarque–Bera still rejects ($p = 0.001$), while Lilliefors ($p = 0.09$) and Anderson–Darling do not. The disagreement is a matter of power: Jarque–Bera targets skewness and kurtosis, precisely where annual returns deviate ($\hat{S} = -0.96$, $\hat{K} = 4.90$), whereas the other two compare the whole distribution function using only 45 points. Two reasons explain why convergence is slow: returns are not independent (their volatility is clustered, stylized fact 6), and with a tail index close to 3 the fourth moment that governs the speed of convergence of the kurtosis is infinite.

Conclusion. Aggregational Gaussianity holds as a tendency: returns move towards normality when aggregated, most of the way by the monthly horizon, but normality is not reached at one year.

4.6 Stylized fact 5: returns are not autocorrelated

Standard tests. Most sample autocorrelations of daily, weekly and monthly returns lie inside the Bartlett bands (Figure 12), with a few exceptions (lags 1, 3, 4, 6, 20 and 22 for daily returns). The autocorrelations are small: $\hat{\rho}_1 = -0.032$ daily and -0.081 weekly. Yet the Ljung–Box test rejects H_0: $\rho_1 = \cdots = \rho_{25} = 0$ at the daily and weekly frequencies ($Q_{LB}(25) = 62.6$ and 94.8 against a critical value of 37.7, Table 14). It does not reject at the monthly frequency. At the annual frequency, $Q_{LB}(10) = 20.1$ ($p = 0.028$) rests on 45 observations, and the Box–Pierce version does not reject ($p = 0.065$, Table 31). The complete tables are in Appendix A.

Robust test. The Ljung–Box statistic and the Bartlett bands rely on $\mathrm{Var}(\hat{\rho}_j) = 1/N$, which holds for i.i.d. returns. When the variance is time-varying, large returns occur together, the dispersion of $\hat{\rho}_j$ is larger than $1/N$, and these tests reject too often. Returns are therefore regressed

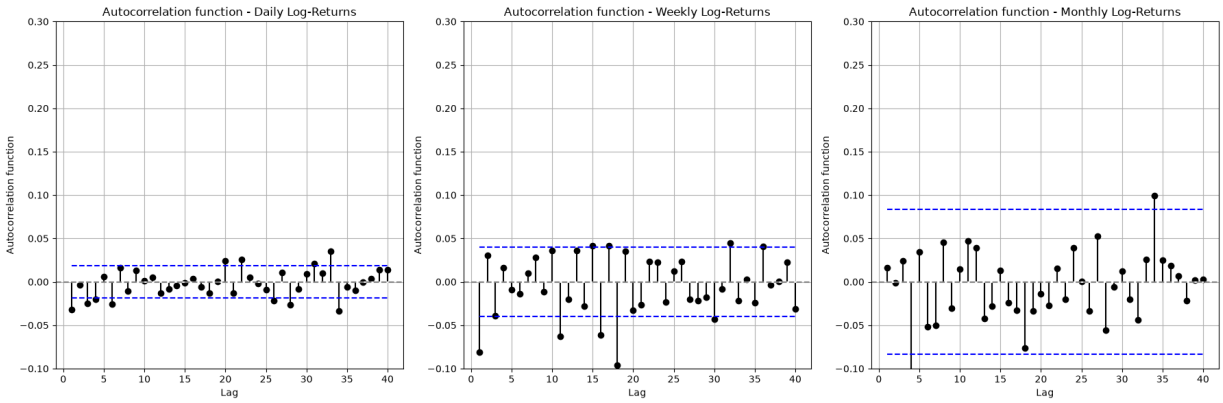


Figure 12: Autocorrelation function of daily, weekly and monthly log-returns (40 lags) with Bartlett bands $\pm 1.96/\sqrt{N}$.

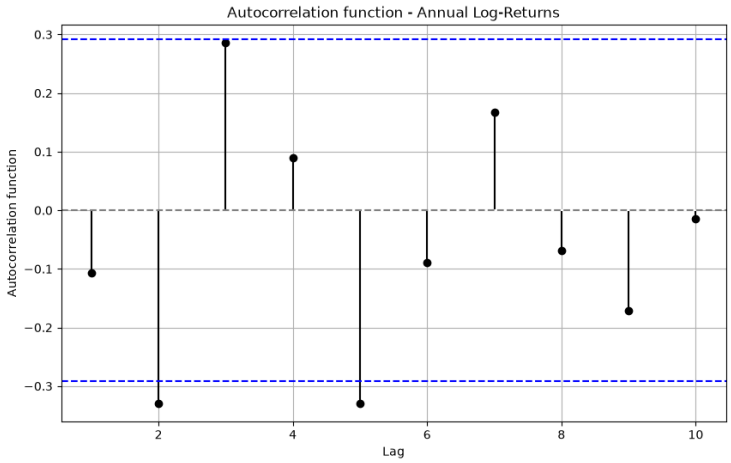


Figure 13: Autocorrelation function of annual log-returns (10 lags, 45 observations) with Bartlett bands.

Table 14: Ljung–Box test by frequency, H_0: $\rho_1 = \cdots = \rho_m = 0$.

	N	$\hat{\rho}_1$	m	$Q_{LB}(m)$	$\chi^2_{0.95}(m)$	p-value
Daily	11,542	−0.032	25	62.6	37.7	<0.001
Weekly	2,389	−0.081	25	94.8	37.7	<0.001
Monthly	549	0.017	25	32.1	37.7	0.154
Annual	45	−0.106	10	20.1	18.3	0.028

on their first five lags,

$$r_t = c + \sum_{j=1}^{5} \phi_j \, r_{t-j} + u_t,$$

and H_0: $\phi_1 = \cdots = \phi_5 = 0$ is tested with the F-statistic computed under both the OLS and the White covariance matrices (Table 15).

Results.

- **Daily and weekly returns.** With White standard errors, the t-statistic of the first lag falls from −3.53 to −1.53 (daily) and from −3.76 to −1.78 (weekly), and the joint test no longer rejects ($p = 0.43$ and 0.46). The Ljung–Box rejections were produced by heteroskedasticity. In any case the predictable share of the variance is negligible ($R^2 = 0.2\%$ and 0.8%).
- **Monthly returns.** The Ljung–Box test at 25 lags does not reject, but the fourth lag ($\hat{\phi}_4 = -0.171$) remains significant with White standard errors ($t = -3.57$; joint $p = 0.024$).

Table 15: Regression of log-returns on their first five lags: coefficients, t-statistics under the OLS and White covariance matrices, and global F-test H_0: $\phi_1 = \cdots = \phi_5 = 0$.

	Daily			Weekly			Monthly		
	$\hat{\phi}_j$	t_{OLS}	t_{White}	$\hat{\phi}_j$	t_{OLS}	t_{White}	$\hat{\phi}_j$	t_{OLS}	t_{White}
Constant	0.0005	2.61	2.54	0.0026	2.66	2.46	0.0110	2.77	2.50
r_{t-1}	−0.0329	−3.53	−1.53	−0.0771	−3.76	−1.78	0.0275	0.64	0.47
r_{t-2}	−0.0054	−0.58	−0.27	0.0220	1.07	0.67	−0.0037	−0.09	−0.06
r_{t-3}	−0.0262	−2.81	−1.43	−0.0338	−1.65	−0.98	0.0273	0.64	0.60
r_{t-4}	−0.0216	−2.32	−1.16	0.0091	0.44	0.32	−0.1712	−4.03	−3.57
r_{t-5}	0.0044	0.48	0.24	−0.0055	−0.27	−0.21	0.0425	0.99	0.88
F-test, all lags		5.08	0.98		4.03	0.93		3.51	2.62
p-value		<0.001	0.426		0.001	0.460		0.004	0.024
R^2	0.2%			0.8%			3.2%		

It is a single isolated lag out of the 25 examined, with no counterpart at the neighbouring lags ($\hat{\rho}_3 = 0.024$, $\hat{\rho}_5 = 0.035$) and no obvious economic interpretation, and the regression explains only 3.2% of the variance. It is reported as a deviation from the stylized fact rather than as exploitable predictability.

Conclusion. Returns are not autocorrelated at the daily and weekly frequencies once heteroskedasticity is accounted for. At the monthly frequency one isolated negative autocorrelation, at lag 4, survives.

4.7 Stylized fact 6: volatility clustering and long memory

Graphical evidence. Large returns of either sign occur in clusters (Figure 14), notably in 1987, 1990–1991, 1998–2002, 2008–2009 and 2020. Since r_t^2 and $|r_t|$ measure the size of a return irrespective of its sign, clustering appears as positive autocorrelation of squared and absolute returns (Figures 15 and 16).

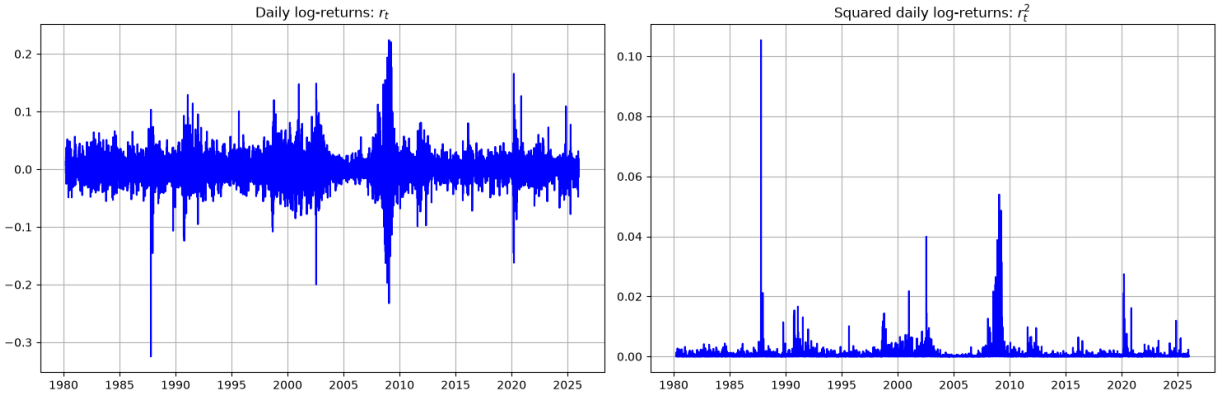


Figure 14: Daily log-returns r_t (left) and squared daily log-returns r_t^2 (right).

Tests. The first 25 autocorrelations of the squared daily returns are all positive, between 0.088 and 0.258, i.e. 5 to 14 times the Bartlett half-width of 0.018. The Ljung–Box statistic is 770 at lag 1 and 5,939 at lag 25, against critical values of 3.84 and 37.7 (Table 32 in the appendix). Engle's ARCH test regresses r_t^2 on m of its own lags and uses $NR^2 \sim \chi^2(m)$ under H_0 of no conditional heteroskedasticity. Unlike the Ljung–Box test, it is explicitly a test of an ARCH structure in the variance. It rejects overwhelmingly at every number of lags (Table 16). Absolute returns, which are less sensitive to extreme observations, are even more persistent: their daily autocorrelations are still about 0.20 at lag 40. Contrary to stylized fact 5, the size of the effect puts the conclusion

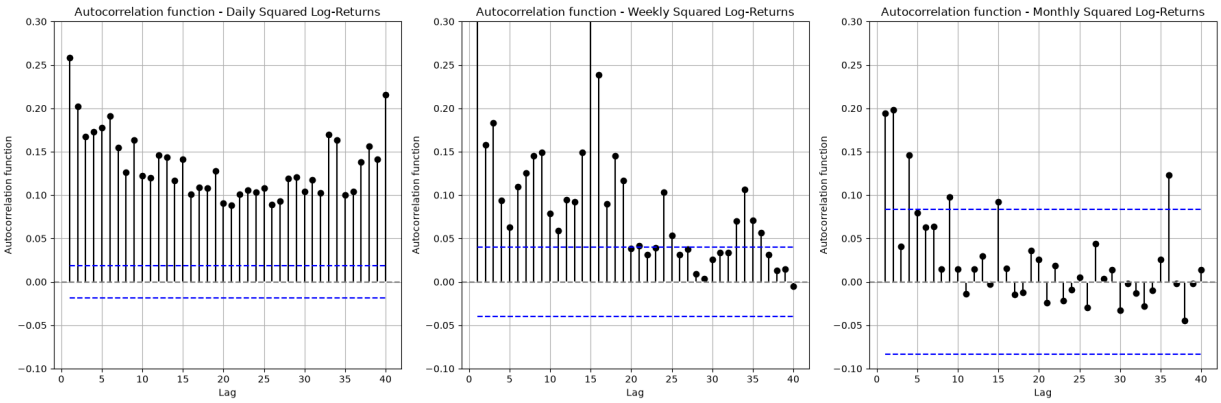


Figure 15: Autocorrelation function of squared daily, weekly and monthly log-returns (40 lags) with Bartlett bands.

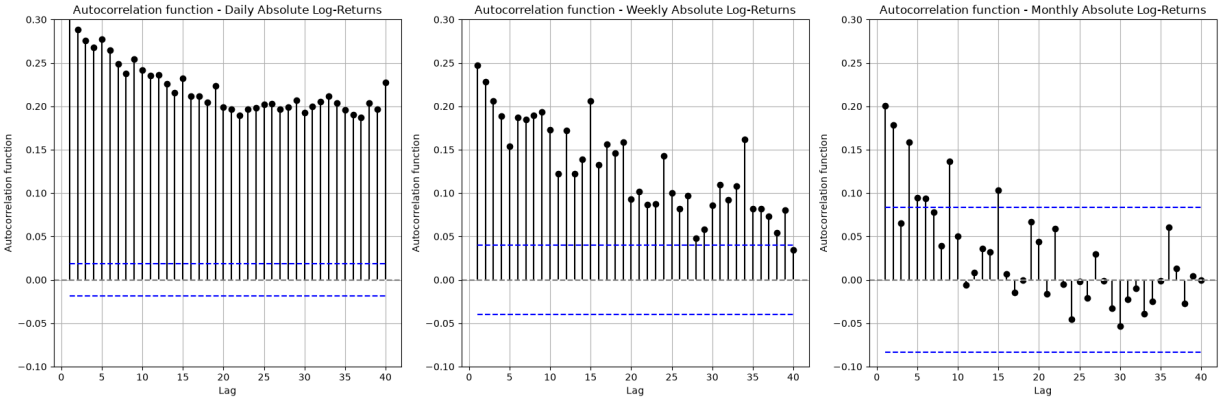


Figure 16: Autocorrelation function of absolute daily, weekly and monthly log-returns (40 lags) with Bartlett bands.

beyond any discussion of the calibration of the tests.

Table 16: Engle's ARCH-LM test on daily log-returns, H_0: no conditional heteroskedasticity.

Number of lags m	1	5	10	25
LM statistic NR^2	770.2	1,280.4	1,464.0	1,581.2
5% critical value	3.84	11.07	18.31	37.65
p-value	<0.001	<0.001	<0.001	<0.001

Table 17: Autocorrelation of squared log-returns by frequency.

Lag	Daily	Weekly	Monthly
1	0.258	0.378	0.195
5	0.178	0.063	0.080
10	0.122	0.079	0.015
20	0.091	0.038	0.025
40	0.215	−0.005	0.014
Bartlett half-width	0.018	0.040	0.084

Sampling frequency. Persistence weakens when the sampling interval increases (Table 17). Daily autocorrelations are still five times the band at lag 20; weekly ones fall inside the band from about lag 20; monthly ones are significant at lag 1 only (0.195 against 0.084). The first weekly

Table 18: Autocorrelation of squared daily log-returns by sub-period.

Lag	1980–1991	1991–2003	2003–2014	2014–2025
1	0.048	0.245	0.353	0.427
5	0.074	0.125	0.227	0.290
20	0.010	0.057	0.138	0.084
60	0.005	0.034	0.091	0.037
Bartlett half-width	0.036	0.036	0.036	0.036

autocorrelation (0.378) is however larger than the daily one, so it is the persistence, not the first lag, that declines.

Sub-periods. Clustering is present in all four sub-periods (Table 18): strongly in the last three ($\hat{\rho}_1$ from 0.25 to 0.43), weakly in 1980–1991 ($\hat{\rho}_1 = 0.048$, $\hat{\rho}_5 = 0.074$, just above the band of 0.037). There, the single observation of 19 October 1987 dominates the variance of r_t^2 and compresses its autocorrelations. The isolated value at lag 40 of the daily series (0.216) illustrates the same sensitivity of the autocorrelation of r_t^2 to a few extreme observations.

GARCH(1,1) model. The tests above detect clustering but do not model it. The conditional variance is specified as

$$h_t = \omega + \alpha\,(r_{t-1} - \bar{r})^2 + \beta\, h_{t-1},$$

in which α measures the reaction of tomorrow's variance to today's shock and $\alpha+\beta$ the persistence of that reaction. The parameters are estimated by maximising the Gaussian log-likelihood $-\frac{1}{2}\sum_t \left(\ln h_t + (r_t - \bar{r})^2/h_t\right)$, written and optimised directly (Nelder–Mead, with $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$), on returns in percent. If the model captures the clustering, the standardized residuals $z_t = (r_t - \bar{r})/\sqrt{h_t}$ must show none: the Ljung–Box statistic on z_t^2 is the test of interest, next to the same statistic on r_t^2.

Table 19: GARCH(1,1) estimated by maximum likelihood on daily log-returns in percent, and Ljung–Box test before and after filtering (critical value 37.7).

Parameter	Estimate	Diagnostic	Value
$\hat{\omega}$ ($\%^2$)	0.0459	Ljung–Box $Q_{LB}(25)$ on r_t^2	5,939.2
$\hat{\alpha}$	0.0831	p-value	<0.001
$\hat{\beta}$	0.9094	Ljung–Box $Q_{LB}(25)$ on z_t^2	25.7
$\hat{\alpha} + \hat{\beta}$	0.9926	p-value	0.426
Half-life $\ln 0.5/\ln(\hat{\alpha} + \hat{\beta})$ (days)	92.8	Kurtosis of z_t	10.19
Implied unconditional st. dev. (%)	2.48	Sample st. dev. (%)	2.20

Results. The persistence $\hat{\alpha} + \hat{\beta} = 0.993$ quantifies the long memory of volatility: a shock loses half of its effect only after 93 trading days, more than four months (Table 19). It is below one, so the model is covariance-stationary, with a finite unconditional variance $\omega/(1 - \alpha - \beta)$. The implied unconditional standard deviation is 2.48% against 2.20% in the sample; so close to one, this quantity is imprecisely estimated. Clustering and a constant unconditional variance are therefore compatible, as argued for stylized fact 1. The conditional volatility (Figure 17) peaks at about 10% per day in late 2008 and returns to about 1–2% in calm periods.

After filtering, the Ljung–Box statistic on the squared residuals falls from 5,939 to 25.7, below the critical value ($p = 0.43$): a single GARCH(1,1) accounts for all the autocorrelation of squared returns. The kurtosis of z_t (10.2) is lower than that of the returns (18.3) but far above 3. Clustering therefore explains part of the heavy tails of stylized fact 3, not all of them: the conditional distribution itself is heavy-tailed.

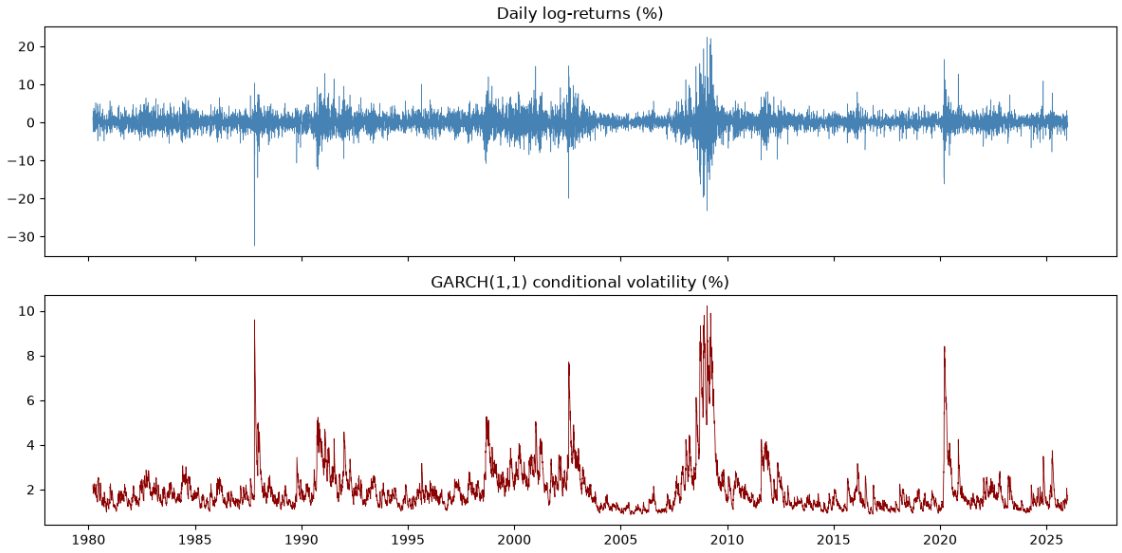


Figure 17: Daily log-returns (top) and GARCH(1,1) conditional volatility $\sqrt{h_t}$ (bottom), in percent.

Conclusion. Volatility clustering is verified, with a persistence that decreases with the sampling interval, and is fully described by a GARCH(1,1) with persistence 0.993.

4.8 Stylized fact 7: leverage effect

The leverage effect refers to the negative relation between past returns and current volatility: a fall in the stock price raises the firm's financial leverage, and hence the volatility of its equity. Since the 1987 crash dominates squared returns in the first sub-period, the main estimates use 1988–2025 ($N = 9{,}582$), with the full sample as a robustness check.

Standard test. The cross-correlations $\hat{\rho}_j = \mathrm{corr}(r_{t-j}, r_t^2)$ are tested with $z = \hat{\rho}_j\sqrt{N}$: one-sided (H_1: $\rho_j < 0$) for $j > 0$, two-sided for $j \leq 0$. On 1988–2025, they are negative for nine of the first ten lags (-0.084, -0.068, -0.057 at $j = 1, 2, 3$; band ± 0.020, Figure 18 and Table 20), and the joint statistic $Q = N\sum_{j=1}^{10}\hat{\rho}_j^2 = 206.5$ far exceeds the $\chi^2(10)$ critical value of 18.3. However, the same rule also declares significant several cross-correlations at negative lags ($j = -1$: 0.057, $z = 5.5$; $j = -9$: 0.059, $z = 5.8$), where no leverage effect can exist, since future returns would explain today's volatility. The contemporaneous value ($j = 0$: $+0.063$) simply reflects the positive skewness of daily returns once 1987 is excluded. As for stylized fact 5, the variance $1/N$ is too small under volatility clustering, and the standard test is not correctly calibrated here.

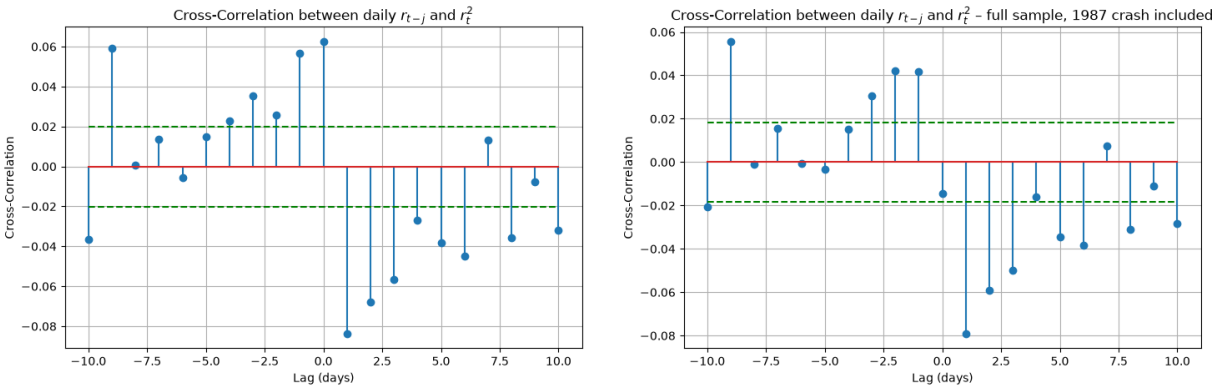


Figure 18: Cross-correlation $\mathrm{corr}(r_{t-j}, r_t^2)$ for $j = -10, \ldots, 10$, daily returns 1988–2025 (left) and full sample including the 1987 crash (right), with bands $\pm 1.96/\sqrt{N}$.

Table 20: Cross-correlations $\hat{\rho}_j = \mathrm{corr}(r_{t-j}, r_t^2)$ of daily log-returns. For $j > 0$ one-sided test H_1: $\rho_j < 0$; for $j \leq 0$ two-sided test. Joint test on $j = 1, \ldots, 10$ (1988–2025): $Q = 206.5$, $\chi^2_{0.95}(10) = 18.3$, $p < 0.001$.

	1988–2025 (N = 9,582)			Full sample
j	$\hat{\rho}_j$	$z = \hat{\rho}_j\sqrt{N}$	p-value	$\hat{\rho}_j$
−10	−0.0363	−3.55	<0.001	−0.0205
−9	0.0590	5.77	<0.001	0.0556
−8	0.0005	0.05	0.960	−0.0010
−7	0.0134	1.31	0.189	0.0155
−6	−0.0055	−0.54	0.591	−0.0005
−5	0.0151	1.47	0.141	−0.0032
−4	0.0230	2.25	0.024	0.0153
−3	0.0355	3.47	0.001	0.0306
−2	0.0259	2.53	0.011	0.0420
−1	0.0566	5.54	<0.001	0.0418
0	0.0625	6.12	<0.001	−0.0146
1	−0.0837	−8.19	<0.001	−0.0791
2	−0.0679	−6.64	<0.001	−0.0590
3	−0.0566	−5.54	<0.001	−0.0501
4	−0.0270	−2.64	0.004	−0.0161
5	−0.0383	−3.74	<0.001	−0.0345
6	−0.0449	−4.39	<0.001	−0.0383
7	0.0134	1.31	0.904	0.0074
8	−0.0358	−3.50	<0.001	−0.0308
9	−0.0077	−0.76	0.225	−0.0108
10	−0.0319	−3.12	0.001	−0.0284

Sensitivity to the 1987 crash. On the full sample, the cross-correlations at positive lags are very similar (−0.079, −0.059, −0.050 at $j = 1, 2, 3$), but the contemporaneous value turns negative (−0.015) and the values at negative lags change: the crash observations dominate r_t^2 and distort the correlations around them.

Robust test. Squared returns are regressed on the ten lagged returns,

$$r_t^2 = c + \sum_{j=1}^{10} \gamma_j \, r_{t-j} + u_t,$$

with White standard errors. A leverage effect implies $\gamma_j < 0$ (Table 21).

Results for JPM. Nine of the ten coefficients are negative in both samples, and the first three remain individually significant with White standard errors ($t = -2.1, -2.4, -2.0$ on 1988–2025). The joint evidence is much weaker than Q suggests: with White standard errors the t-statistics are divided by about four, and the robust F-test rejects at 5% on the full sample ($p = 0.047$) but only at 10% on 1988–2025 ($p = 0.091$). In economic terms, a daily return of −2% raises the next day's expected squared return by $2 \times 7.46 \times 10^{-3} \times 10^{-2} \approx 1.5 \times 10^{-4}$, i.e. by about 28% of its average level (5.3×10^{-4}, the constant of the regression).

Market-level evidence. The VIX, the implied volatility of S&P 500 options, rises sharply in every crisis, when JPM falls (Figure 19). Daily VIX changes are regressed on JPM daily log-returns, first symmetrically, then allowing a different slope for negative and positive returns:

$$\Delta VIX_t = a + \beta \, r_t + e_t, \qquad \Delta VIX_t = a + \beta^- \min(r_t, 0) + \beta^+ \max(r_t, 0) + e_t.$$

Table 21: Regression of squared daily log-returns on ten lagged returns: coefficients ($\times 10^3$), t-statistics under the OLS and White covariance matrices, and global F-test H_0: $\gamma_1 = \cdots = \gamma_{10} = 0$.

	1988–2025			Full sample (with 1987)		
	$\hat{\gamma}_j \times 10^3$	t_{OLS}	t_{White}	$\hat{\gamma}_j \times 10^3$	t_{OLS}	t_{White}
Constant	0.53	27.39	23.64	0.50	26.94	23.80
r_{t-1}	−7.46	−8.70	−2.12	−7.50	−8.85	−2.35
r_{t-2}	−6.61	−7.71	−2.39	−6.08	−7.17	−2.50
r_{t-3}	−5.26	−6.13	−2.00	−4.97	−5.86	−2.14
r_{t-4}	−3.06	−3.56	−1.14	−2.10	−2.48	−0.88
r_{t-5}	−4.04	−4.71	−1.84	−3.76	−4.44	−1.93
r_{t-6}	−4.31	−5.03	−1.57	−3.94	−4.64	−1.65
r_{t-7}	0.35	0.41	0.14	0.06	0.07	0.03
r_{t-8}	−3.25	−3.79	−1.60	−3.04	−3.58	−1.72
r_{t-9}	−1.21	−1.42	−0.45	−1.45	−1.71	−0.60
r_{t-10}	−2.77	−3.24	−1.45	−2.66	−3.14	−1.58
F-test, 10 lags		23.46	1.63		21.78	1.86
p-value		<0.001	0.091		<0.001	0.046
R^2	2.4%			1.9%		

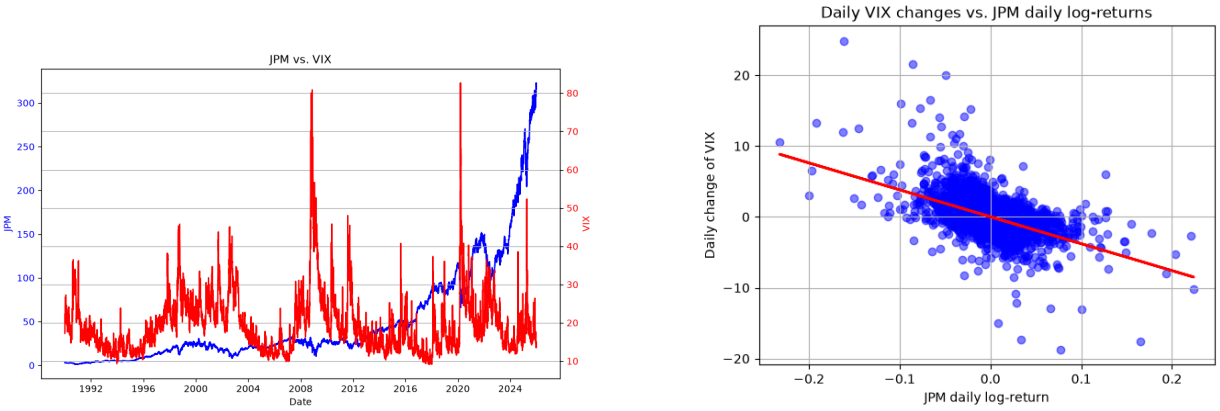


Figure 19: Left: JPM price (blue) and VIX level (red), 1990–2025. Right: daily change of the VIX against the JPM daily log-return, with the fitted regression line.

Table 22: Regressions of the daily change of the VIX on the JPM daily log-return (1990–2025, $N = 9{,}066$; correlation −0.52). The F-test of $\beta^- = \beta^+$ is reported with its p-value.

	Estimate	t_{OLS} / F_{OLS}	t_{White} / F_{White}	R^2
β (all returns)	−38.02	−58.23	−24.91	0.272
β^- (negative returns)	−49.03	−44.45	−16.27	0.284
β^+ (positive returns)	−27.93	−26.76	−12.81	
F-test $\beta^- = \beta^+$		152.0 (<0.001)	24.0 (<0.001)	

The effect is clear at the market level (Table 22). The daily change of the VIX is negatively related to the JPM return (correlation −0.52): a return of −1% is associated with a rise of 0.38 VIX point. The response is asymmetric: −49.0 for negative returns against −27.9 for positive ones, so a fall of JPM raises implied volatility almost twice as much as a rise of the same size lowers it. Both results survive the use of White standard errors, although the t- and F-statistics are divided by 2 to 6. Since the VIX is the implied volatility of the S&P 500, this regression documents the leverage effect of the market, of which JPM is a high-beta component, rather than that of JPM alone.

Conclusion. The leverage effect is verified in sign and for the first lags. Its joint statistical significance in JPM's own returns is borderline once heteroskedasticity is taken into account; at the market level it is strong and asymmetric.

4.9 Summary of the stylized facts

Table 23: Summary of the stylized facts of JPM returns.

Stylized fact	Verdict	Main evidence
1. Stationarity	Holds for mean and autocovariance; compatible for the variance	Homogeneity not rejected for the mean ($p = 0.90$) and autocovariance ($p = 0.25$); variance rejected because of clustering, GARCH persistence < 1
2. Negative skewness	Weekly to monthly horizons only	Daily -0.11 becomes $+0.17$ without October 1987; -0.34 at 5 days, -0.53 at 21 days
3. Heavy tails	Strongly verified	Daily kurtosis 18.3; Hill tail index 2.5–3.0; Student-t $\hat{\nu} = 2.82$
4. Aggregational Gaussianity	Verified as a tendency	Kurtosis 18.3 to 4.9, JB/N 9.74 to 0.31; normality still rejected at one year by Jarque–Bera
5. No autocorrelation	Verified daily and weekly with robust inference	Ljung–Box rejects, robust F-test does not ($p = 0.43$, 0.46); isolated monthly lag 4
6. Volatility clustering	Strongly verified	ARCH-LM 1,581 (25 lags); GARCH persistence 0.993, half-life 93 days
7. Leverage effect	Verified in sign; borderline for JPM alone	9 of 10 lags negative, robust F-test $p = 0.047$–0.091; asymmetric VIX response

5 Factor models: CAPM and Fama–French

5.1 Specification

Monthly JPM excess returns, in percent and net of the one-month Treasury bill rate, are regressed on the Fama–French factors: the excess return of the CRSP value-weighted index (MKT), SMB and HML. The CRSP index return is rebuilt as Mkt$-$RF + RF, and its excess return over RF is used as the market factor. The sample runs from April 1980 to December 2025 ($T = 549$). Three nested models are estimated by OLS:

$$\begin{aligned} \text{Model 0:} \quad & R_t - R_{f,t} = \alpha + \varepsilon_t, \\ \text{CAPM (market model):} \quad & R_t - R_{f,t} = \alpha + \beta_M \, MKT_t + \varepsilon_t, \\ \text{Fama–French:} \quad & R_t - R_{f,t} = \alpha + \beta_M \, MKT_t + \beta_S \, SMB_t + \beta_H \, HML_t + \varepsilon_t. \end{aligned}$$

JPM returns are simple returns, consistent with the factors (Section 5.7).

5.2 Estimates

Table 24 reports the estimates of the three models. The market factor is highly significant in both the CAPM and the Fama–French model, HML is highly significant, SMB is not, and neither of the two factor models leaves a significant intercept. The fitted market model is shown in Figure 20. Inference, tests of restrictions and the economic interpretation of the coefficients follow.

Table 24: OLS estimates of the three models, JPM monthly excess simple returns in percent. OLS standard errors in parentheses, White (HC0) standard errors in brackets. Significance with OLS standard errors: * 10%, ** 5%, *** 1%.

	Model 0	CAPM	Fama–French
α (% per month)	1.124***	0.192	−0.086
	(0.381) [0.381]	(0.304) [0.310]	(0.284) [0.289]
MKT		1.237***	1.345***
		(0.067) [0.078]	(0.065) [0.075]
SMB			0.158
			(0.098) [0.114]
HML			0.871***
			(0.092) [0.104]
R^2	0.000	0.382	0.470
Adjusted R^2	0.000	0.381	0.467
Global F-statistic	—	338.4	161.2
Log-likelihood	−1,980.8	−1,848.6	−1,806.4
Residual skewness	−0.241	0.263	0.194
Residual kurtosis	4.594	5.791	5.697
Residual Jarque–Bera	63.5	184.5	169.8
Durbin–Watson	1.977	2.106	2.102
T	549	549	549

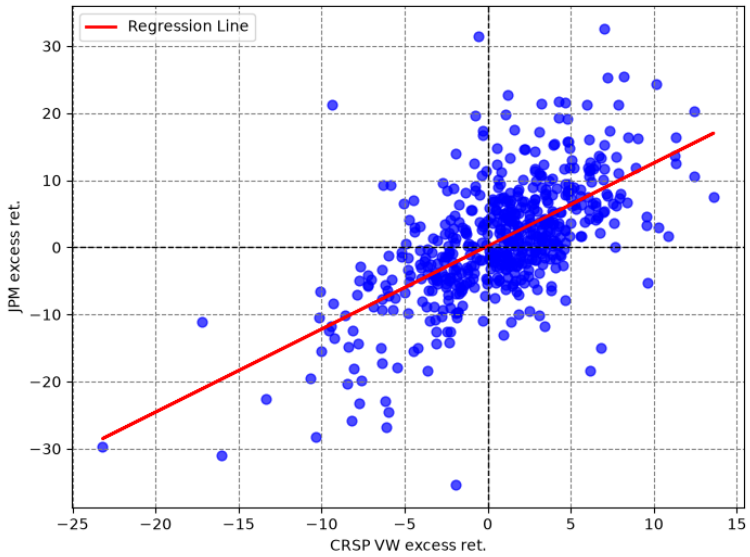


Figure 20: Market model: JPM monthly excess return against the CRSP value-weighted excess return (%), with the fitted regression line $\hat{y} = 0.192 + 1.237\,x$.

5.3 Inference: OLS and White standard errors

The residuals of the three models are far from normal: their kurtosis lies between 4.6 and 5.8, and the Jarque–Bera statistics (63 to 185) far exceed the critical value of 5.99, in line with stylized facts 3 and 4. The White standard errors of the slopes are 14% to 16% larger than the OLS ones (Table 25), a sign of heteroskedasticity; the HC3 correction changes them very little. With $T = 549$, the tests remain valid asymptotically, but White standard errors are reported next to the OLS ones and the conclusions are based on both. The Durbin–Watson statistics (1.98 to 2.11) show no first-order autocorrelation of the residuals.

Table 25: Standard errors under the OLS covariance matrix and White's HC0 and HC3 heteroskedasticity-consistent covariance matrices.

Model		Estimate	s.e. OLS	s.e. White HC0	s.e. White HC3
Model 0	α	1.1237	0.3813	0.3810	0.3817
CAPM	α	0.1920	0.3042	0.3102	0.3118
	MKT	1.2375	0.0673	0.0779	0.0787
Fama–French	α	−0.0859	0.2839	0.2886	0.2919
	MKT	1.3454	0.0651	0.0751	0.0766
	SMB	0.1581	0.0979	0.1138	0.1158
	HML	0.8714	0.0917	0.1043	0.1067

5.4 Tests of restrictions

Table 26: F-tests of linear restrictions under the OLS and White covariance matrices. For a single restriction $F = t^2$.

Hypothesis	F_{OLS}	p-value	F_{White}	p-value
CAPM: $\alpha = 0$	0.40	0.528	0.38	0.536
CAPM: $\beta_M = 1$	12.46	0.001	9.29	0.002
Fama–French: $\alpha = 0$	0.09	0.762	0.09	0.766
Fama–French: $\beta_M = 1$	28.13	<0.001	21.13	<0.001
Fama–French: $\beta_S = \beta_H = 0$	45.20	<0.001	35.76	<0.001

For a stock, the informative null hypothesis on the market beta is $\beta_M = 1$ rather than $\beta_M = 0$: the question is whether the stock is riskier than the market, not whether it moves with it. The restriction $\beta_S = \beta_H = 0$ tests the Fama–French model against the CAPM (Table 26).

5.5 Significance and magnitude of the coefficients

Market beta. In the market model $\hat{\beta}_M = 1.24$ ($t = 18.4$): a 1% excess return of the market is associated with a 1.24% excess return of JPM. In a month in which the market loses 5% in excess of the risk-free rate, the model predicts $0.19 + 1.24 \times (-5) \approx -6.0\%$ for JPM. The hypothesis $\beta_M = 1$ gives $t = (1.237 - 1)/0.067 = 3.53$, i.e. $F = t^2 = 12.5$ (9.3 with White standard errors, $p = 0.002$). It is rejected: JPM is significantly riskier than the market, as expected from a highly levered bank whose earnings are strongly pro-cyclical. The fitted line of Figure 20 is accordingly steeper than the 45-degree line and passes very close to the origin. The beta rises to 1.35 once SMB and HML are included, because HML is negatively correlated with the market over the sample (the CAPM beta absorbs part of the value exposure), and $\beta_M = 1$ is rejected even more strongly ($F = 28.1$).

Value factor. The HML loading is large and highly significant: 0.87, with $t = 9.5$ (8.4 with White standard errors). Holding the market fixed, a 1% return of the value-minus-growth portfolio is associated with a 0.87% return of JPM. JPM behaves like a value stock, which is typical of banks, whose book-to-market ratios are high.

Size factor. The SMB loading is small (0.16) and not significant, even at 10% ($t = 1.62$; 1.39 with White standard errors). JPM being one of the largest US companies, no positive exposure to the size factor was expected.

Joint test. SMB and HML are jointly highly significant, entirely because of HML:

$$F = \frac{(R^2_{FF} - R^2_{CAPM})/2}{(1 - R^2_{FF})/(T-4)} = \frac{(0.4701 - 0.3822)/2}{0.5299/545} = 45.2,$$

against a 5% critical value of 3.0 (35.8 with White standard errors).

Goodness of fit. Model 0 explains nothing by construction ($R^2 = 0$): it only estimates the mean. The market factor explains 38.2% of the variance of JPM excess returns; adding SMB and HML raises the adjusted R^2 from 0.381 to 0.467, an improvement confirmed by the joint F-test. The global F-statistics (338.4 and 161.2) reject the hypothesis that all slopes are zero in both models. About half of the variance (53%) remains idiosyncratic, which is usual for a single stock: this firm-specific risk can be diversified away and, according to theory, earns no premium.

5.6 The intercepts

Model 0. The constant is the average excess return of JPM: 1.12% per month, about 13.5% per year, significant at the 1% level ($t = 2.95$). Its positive sign is what theory predicts for a risky asset.

CAPM. The CAPM states that the expected excess return of an asset is entirely explained by its beta, $E(R_i - R_f) = \beta_i\, E(R_M - R_f)$, hence $\alpha = 0$ in the time-series regression. The estimate is +0.19% per month and far from significant ($t = 0.63$, $p = 0.53$; $p = 0.54$ with White standard errors): the data are compatible with the CAPM. Since the regression line passes through the sample means, the average excess return decomposes as

$$\overline{R - R_f} = \hat{\alpha} + \hat{\beta}_M\, \overline{MKT}: \qquad 1.124 = 0.192 + 1.237 \times 0.753.$$

The market premium over the sample is 0.75% per month, JPM's beta entitles it to 0.93%, and it earned 1.12%. In the language of portfolio theory, $\hat{\alpha}$ is Jensen's alpha: JPM lies slightly above the security market line, by about 2.3% per year, but the distance is well within sampling error.

Fama–French model. In a multi-factor model of the APT type, the absence of arbitrage implies that expected excess returns are linear in the loadings on the systematic factors, $E(R_i - R_f) = \sum_k \beta_{ik}\lambda_k$. When the factors are themselves excess returns or long-short portfolios, as here, their premia λ_k are their means, and the intercept of the time-series regression should again be zero. A significant alpha would mean that JPM offers a return not explained by its exposure to the three sources of risk. The estimate is −0.09% per month ($t = -0.30$, $p = 0.76$): given its market and value exposures, JPM earned what the factor premia imply.

5.7 Simple returns versus log-returns

The Fama–French factors are simple returns, so the regressions use simple returns for JPM as well: $R_t - R_{f,t}$ is then the return on a self-financing position, which is the quantity the CAPM and the APT price, whereas $\ln(1 + R_t) - R_{f,t}$ subtracts an arithmetic rate from a logarithmic one. The stylized facts use log-returns, so the three models are re-estimated with $\ln(P_t/P_{t-1})$ on the left-hand side for comparison (Table 27). Over a month, the difference between the two returns is of second order, $\ln(1 + R_t) \approx R_t - \frac{1}{2}R_t^2$, but its mean is not: it lowers the average JPM return by about $\sigma^2/2$, here $9.04^2/200 \approx 0.41\%$ per month, the same order of magnitude as the estimated intercepts.

The three intercepts shift by 0.41 to 0.44 percentage points, exactly the predicted amount, while the slopes and the R^2 are almost unchanged. With log-returns, the average excess return is only significant at 10% ($t = 1.85$), the CAPM alpha becomes negative (−0.24, $t = -0.80$), and the Fama–French alpha appears negative and significant at 10% (−0.52, $t = -1.83$). That marginally

Table 27: Estimates with JPM simple returns and with JPM log-returns (OLS t-statistics). The last column is the difference simple minus log.

	Simple	t	Log	t	Difference
Model 0: α	1.124	2.95	0.713	1.85	0.410
CAPM: α	0.192	0.63	−0.244	−0.80	0.436
CAPM: MKT	1.237	18.40	1.272	18.84	−0.034
CAPM: R^2	0.382		0.394		−0.011
Fama–French: α	−0.086	−0.30	−0.521	−1.83	0.435
Fama–French: MKT	1.345	20.66	1.378	21.08	−0.032
Fama–French: SMB	0.158	1.62	0.172	1.75	−0.014
Fama–French: HML	0.871	9.51	0.872	9.48	0.000
Fama–French: R^2	0.470		0.479		−0.009

significant under-performance is an artefact of mixing the two return conventions: it disappears when JPM and the factors are measured consistently.

Conclusion. Without factors, the intercept is a positive risk premium, significant at 1%. Once the market factor, or the three factors, are included, the intercept is not significantly different from zero, and practically zero in the three-factor model. The results are compatible with the CAPM and with the multi-factor view, the three-factor model providing the better description of JPM's risk exposures: a market beta well above one and a strong value tilt.

6 Conclusion

JPMorgan Chase returns display most of the classical stylized facts, but the strength of the evidence depends heavily on whether inference accounts for the very features being documented. Heavy tails and volatility clustering are the dominant properties of the data. The tail index is close to 3, so the fourth moment is infinite, and volatility follows a highly persistent but stationary GARCH process. These two properties in turn invalidate the i.i.d. assumptions behind several standard tests. Once robust inference is used, the apparent autocorrelation of daily and weekly returns disappears, and the leverage effect in JPM's own returns becomes borderline, although it remains strong and asymmetric at the market level. The negative skewness usually reported for daily returns is, for JPM, the product of a single week in October 1987; it is robust only at weekly to monthly horizons.

In the cross-section of risk, JPM is a high-beta value stock: its market beta is significantly above one, its HML loading is large, and it has no size exposure. Neither the CAPM nor the Fama–French model leaves a significant alpha, provided that JPM and the factors are measured with the same return convention.

7 Limitations and extensions

- **Single asset.** All conclusions concern one stock. Before 2000, the price history is that of JPMorgan Chase's predecessor entities, and the business changed through several major mergers over the sample.
- **Normal-based standard errors.** The skewness and kurtosis z-tests use standard errors that are valid under normality only. Under heavy tails they overstate significance; bootstrap standard errors would give a more reliable measure of uncertainty.

- **GARCH specification.** The GARCH(1,1) uses a Gaussian likelihood, while the standardized residuals keep a kurtosis of 10.2. A Student-t likelihood, or an asymmetric specification such as GJR-GARCH or EGARCH that models the leverage effect directly, is a natural extension.
- **Market-level volatility.** The VIX measures S&P 500 implied volatility, so the VIX regressions capture the market's leverage effect rather than JPM's. JPM option-implied volatility would isolate the firm-level effect.
- **Constant loadings.** Factor loadings are assumed constant over 45 years. Rolling or regime-dependent betas, and additional factors (momentum, profitability, investment), are left for further work.
- **Data revisions.** Yahoo Finance may revise historical prices, so re-running the analysis can produce slightly different figures.

A Complete autocorrelation tables

Each table reports the sample autocorrelation $\hat{\rho}_j$, the statistic $z = \hat{\rho}_j\sqrt{N}$, the Box–Pierce statistic $Q_{BP}(j) = N\sum_{i\le j}\hat{\rho}_i^2$ and the Ljung–Box statistic $Q_{LB}(j)$ for H_0: $\rho_1 = \cdots = \rho_j = 0$, with the 5% critical value of the $\chi^2(j)$ distribution.

Table 28: Daily log-returns ($N = 11{,}542$).

Lag j	$\hat{\rho}_j$	$z = \hat{\rho}_j\sqrt{N}$	$Q_{BP}(j)$	p-value	$Q_{LB}(j)$	p-value	$\chi^2_{0.95}(j)$
1	−0.032	−3.47	12.05	0.001	12.05	0.001	3.84
2	−0.003	−0.37	12.19	0.002	12.19	0.002	5.99
3	−0.025	−2.71	19.55	<0.001	19.55	<0.001	7.82
4	−0.020	−2.16	24.20	<0.001	24.20	<0.001	9.49
5	0.006	0.65	24.61	<0.001	24.62	<0.001	11.07
6	−0.026	−2.80	32.46	<0.001	32.47	<0.001	12.59
7	0.016	1.76	35.54	<0.001	35.56	<0.001	14.07
8	−0.011	−1.15	36.87	<0.001	36.88	<0.001	15.51
9	0.013	1.40	38.82	<0.001	38.84	<0.001	16.92
10	0.001	0.10	38.83	<0.001	38.85	<0.001	18.31
11	0.005	0.53	39.11	<0.001	39.13	<0.001	19.68
12	−0.013	−1.40	41.06	<0.001	41.08	<0.001	21.03
13	−0.008	−0.91	41.88	<0.001	41.90	<0.001	22.36
14	−0.004	−0.46	42.09	<0.001	42.11	<0.001	23.68
15	−0.001	−0.12	42.10	<0.001	42.13	<0.001	25.00
16	0.004	0.42	42.28	<0.001	42.31	<0.001	26.30
17	−0.006	−0.62	42.67	0.001	42.69	0.001	27.59
18	−0.013	−1.44	44.74	<0.001	44.77	<0.001	28.87
19	0.001	0.08	44.75	0.001	44.78	0.001	30.14
20	0.024	2.56	51.32	<0.001	51.36	<0.001	31.41
21	−0.013	−1.41	53.30	<0.001	53.35	<0.001	32.67
22	0.026	2.80	61.16	<0.001	61.23	<0.001	33.92
23	0.005	0.56	61.48	<0.001	61.55	<0.001	35.17
24	−0.002	−0.18	61.51	<0.001	61.58	<0.001	36.41
25	−0.009	−0.99	62.49	<0.001	62.55	<0.001	37.65

Table 29: Weekly log-returns ($N = 2{,}389$).

Lag j	$\hat{\rho}_j$	$z = \hat{\rho}_j\sqrt{N}$	$Q_{BP}(j)$	p-value	$Q_{LB}(j)$	p-value	$\chi^2_{0.95}(j)$
1	−0.081	−3.95	15.60	<0.001	15.62	<0.001	3.84
2	0.031	1.51	17.87	<0.001	17.89	<0.001	5.99
3	−0.039	−1.90	21.47	<0.001	21.50	<0.001	7.82
4	0.016	0.79	22.09	<0.001	22.12	<0.001	9.49
5	−0.009	−0.46	22.30	<0.001	22.33	<0.001	11.07
6	−0.014	−0.67	22.74	0.001	22.78	0.001	12.59
7	0.010	0.48	22.97	0.002	23.01	0.002	14.07
8	0.028	1.38	24.89	0.002	24.93	0.002	15.51
9	−0.012	−0.57	25.21	0.003	25.26	0.003	16.92
10	0.036	1.75	28.27	0.002	28.33	0.002	18.31
11	−0.063	−3.07	37.71	<0.001	37.82	<0.001	19.68
12	−0.020	−0.97	38.65	<0.001	38.77	<0.001	21.03
13	0.036	1.78	41.81	<0.001	41.95	<0.001	22.36
14	−0.028	−1.38	43.70	<0.001	43.85	<0.001	23.68
15	0.041	2.03	47.81	<0.001	47.99	<0.001	25.00
16	−0.061	−2.98	56.67	<0.001	56.92	<0.001	26.30
17	0.041	2.02	60.73	<0.001	61.01	<0.001	27.59
18	−0.096	−4.71	82.89	<0.001	83.36	<0.001	28.87
19	0.035	1.71	85.82	<0.001	86.31	<0.001	30.14
20	−0.033	−1.60	88.39	<0.001	88.91	<0.001	31.41
21	−0.026	−1.28	90.02	<0.001	90.55	<0.001	32.67
22	0.023	1.14	91.31	<0.001	91.86	<0.001	33.92
23	0.023	1.10	92.53	<0.001	93.09	<0.001	35.17
24	−0.023	−1.13	93.81	<0.001	94.38	<0.001	36.41
25	0.012	0.61	94.18	<0.001	94.75	<0.001	37.65

Table 30: Monthly log-returns ($N = 549$).

Lag j	$\hat{\rho}_j$	$z = \hat{\rho}_j\sqrt{N}$	$Q_{BP}(j)$	p-value	$Q_{LB}(j)$	p-value	$\chi^2_{0.95}(j)$
1	0.017	0.39	0.15	0.697	0.15	0.696	3.84
2	−0.001	−0.03	0.15	0.927	0.15	0.926	5.99
3	0.024	0.57	0.47	0.925	0.48	0.924	7.82
4	−0.170	−3.97	16.25	0.003	16.43	0.002	9.49
5	0.035	0.81	16.92	0.005	17.10	0.004	11.07
6	−0.052	−1.21	18.39	0.005	18.60	0.005	12.59
7	−0.050	−1.17	19.76	0.006	19.99	0.006	14.07
8	0.045	1.06	20.89	0.007	21.14	0.007	15.51
9	−0.031	−0.72	21.41	0.011	21.67	0.010	16.92
10	0.014	0.34	21.52	0.018	21.79	0.016	18.31
11	0.047	1.10	22.73	0.019	23.02	0.018	19.68
12	0.039	0.92	23.56	0.023	23.88	0.021	21.03
13	−0.042	−0.99	24.55	0.026	24.89	0.024	22.36
14	−0.028	−0.66	24.99	0.035	25.34	0.031	23.68
15	0.013	0.31	25.08	0.049	25.44	0.044	25.00
16	−0.024	−0.57	25.41	0.063	25.77	0.057	26.30
17	−0.033	−0.78	26.01	0.074	26.40	0.067	27.59
18	−0.076	−1.79	29.22	0.046	29.73	0.040	28.87
19	−0.033	−0.78	29.84	0.054	30.37	0.047	30.14
20	−0.014	−0.33	29.94	0.071	30.48	0.062	31.41
21	−0.027	−0.63	30.34	0.085	30.90	0.075	32.67
22	0.015	0.36	30.47	0.108	31.03	0.095	33.92
23	−0.020	−0.48	30.70	0.130	31.27	0.116	35.17
24	0.039	0.91	31.53	0.139	32.15	0.123	36.41
25	0.001	0.02	31.53	0.172	32.15	0.154	37.65

Table 31: Annual log-returns ($N = 45$).

Lag j	$\hat{\rho}_j$	$z = \hat{\rho}_j\sqrt{N}$	$Q_{BP}(j)$	p-value	$Q_{LB}(j)$	p-value	$\chi^2_{0.95}(j)$
1	−0.106	−0.71	0.51	0.476	0.54	0.461	3.84
2	−0.329	−2.21	5.38	0.068	5.87	0.053	5.99
3	0.286	1.92	9.07	0.028	10.00	0.019	7.82
4	0.090	0.60	9.43	0.051	10.41	0.034	9.49
5	−0.330	−2.21	14.32	0.014	16.15	0.006	11.07
6	−0.089	−0.60	14.67	0.023	16.58	0.011	12.59
7	0.167	1.12	15.92	0.026	18.13	0.011	14.07
8	−0.069	−0.46	16.14	0.040	18.40	0.018	15.51
9	−0.172	−1.15	17.46	0.042	20.13	0.017	16.92
10	−0.014	−0.09	17.47	0.065	20.14	0.028	18.31

Table 32: Squared daily log-returns ($N = 11{,}542$).

Lag j	$\hat{\rho}_j$	$z = \hat{\rho}_j\sqrt{N}$	$Q_{BP}(j)$	p-value	$Q_{LB}(j)$	p-value	$\chi^2_{0.95}(j)$
1	0.258	27.75	770.22	<0.001	770.42	<0.001	3.84
2	0.202	21.71	1,241.76	<0.001	1,242.12	<0.001	5.99
3	0.167	17.97	1,564.82	<0.001	1,565.32	<0.001	7.82
4	0.173	18.56	1,909.18	<0.001	1,909.86	<0.001	9.49
5	0.178	19.08	2,273.19	<0.001	2,274.09	<0.001	11.07
6	0.191	20.49	2,692.94	<0.001	2,694.14	<0.001	12.59
7	0.155	16.62	2,969.09	<0.001	2,970.51	<0.001	14.07
8	0.126	13.53	3,152.19	<0.001	3,153.76	<0.001	15.51
9	0.163	17.55	3,460.24	<0.001	3,462.10	<0.001	16.92
10	0.122	13.10	3,631.84	<0.001	3,633.88	<0.001	18.31
11	0.120	12.87	3,797.33	<0.001	3,799.56	<0.001	19.68
12	0.146	15.72	4,044.42	<0.001	4,046.95	<0.001	21.03
13	0.143	15.39	4,281.23	<0.001	4,284.07	<0.001	22.36
14	0.117	12.54	4,438.51	<0.001	4,441.56	<0.001	23.68
15	0.142	15.21	4,669.75	<0.001	4,673.15	<0.001	25.00
16	0.101	10.82	4,786.85	<0.001	4,790.43	<0.001	26.30
17	0.109	11.69	4,923.38	<0.001	4,927.18	<0.001	27.59
18	0.108	11.57	5,057.34	<0.001	5,061.38	<0.001	28.87
19	0.128	13.74	5,246.13	<0.001	5,250.51	<0.001	30.14
20	0.091	9.76	5,341.32	<0.001	5,345.88	<0.001	31.41
21	0.088	9.44	5,430.50	<0.001	5,435.24	<0.001	32.67
22	0.101	10.83	5,547.80	<0.001	5,552.78	<0.001	33.92
23	0.105	11.32	5,675.94	<0.001	5,681.20	<0.001	35.17
24	0.103	11.08	5,798.73	<0.001	5,804.27	<0.001	36.41
25	0.108	11.60	5,933.33	<0.001	5,939.18	<0.001	37.65

B Reproducibility

The analysis is contained in a single Python notebook. It relies on `pandas` and `numpy` for data handling, `yfinance` to download JPM and VIX prices, `scipy` for distributions, normality tests, the Student-t fit and the GARCH optimisation, `statsmodels` for OLS, robust covariance matrices, the Lilliefors, Ljung–Box and ARCH tests, and `matplotlib` for the figures. The GARCH(1,1) likelihood is coded directly rather than taken from a dedicated package. The Fama–French monthly factors are read from the CSV file of the Kenneth R. French Data Library, which must be placed in the same folder as the notebook.